K. J. Somaiya School of Engineering, Mumbai-77

Department of Sciences and Humanities

Course Name:	Basic of Electrical Engineering Laboratory (BEEL) 316U06L102	Semester:	I
Date of Performance:	/ / 20--	Batch No:	F1
Faculty Name:		Roll No:	7
Faculty Sign&Date:		Grade/Marks:	

Experiment No: 5

Title: Mobile Battery Charger

Aim and Objective of the Experiment:

- To understand the working of Mobile Battery Charging Circuit
- To implement the circuit of Mobile Battery charger on Breadboard and observe the waveforms at various points (Input and output Waveforms for Bridge Rectifier) and measure the output voltage

Requirements:

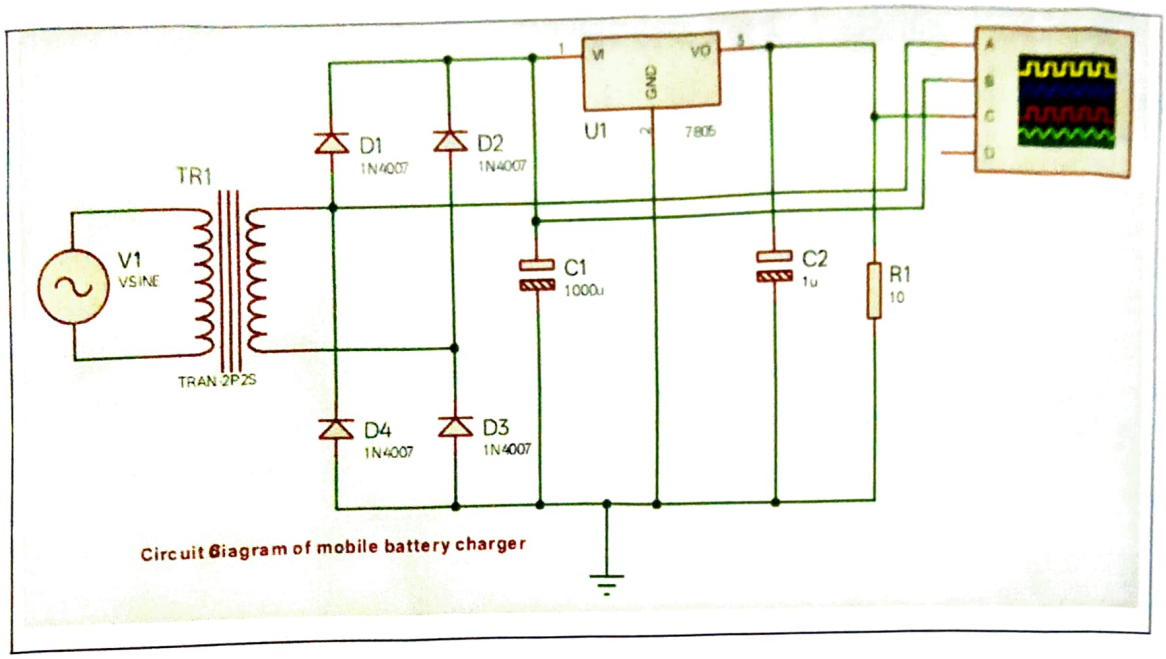
Step-down Transformer (+/= 12 v), Diodes(1N4007), voltage regulator IC 7805, Resistor, Capacitors((1000µF, 100µF, 10µF,0.01 µF), CRO, Digital Multimeter (DMM), breadboard, connecting wires, Micro USB cable, etc.

Theory:

A mobile battery charger converts the supply into a regulated DC voltage suitable for charging a mobile battery. In this circuit, a step down transformer first reduces the AC voltage to a lower value. The

reduced AC is applied to a bridge rectifier made using four IN4007 diodes, which converts AC into pulsating DC. A capacitor is used to smooth the pulsating DC ~~A capacitor is used to smo~~ and reduce the riddles. Finally the IC 7805 voltage regulator provides constant regulated DC output of approx 5V, which can be supplied to the mobile through a USB cable. This input & output waveforms of the bridge rectifier can be observed using a CRO.

Circuit Diagram/ Block Diagram:

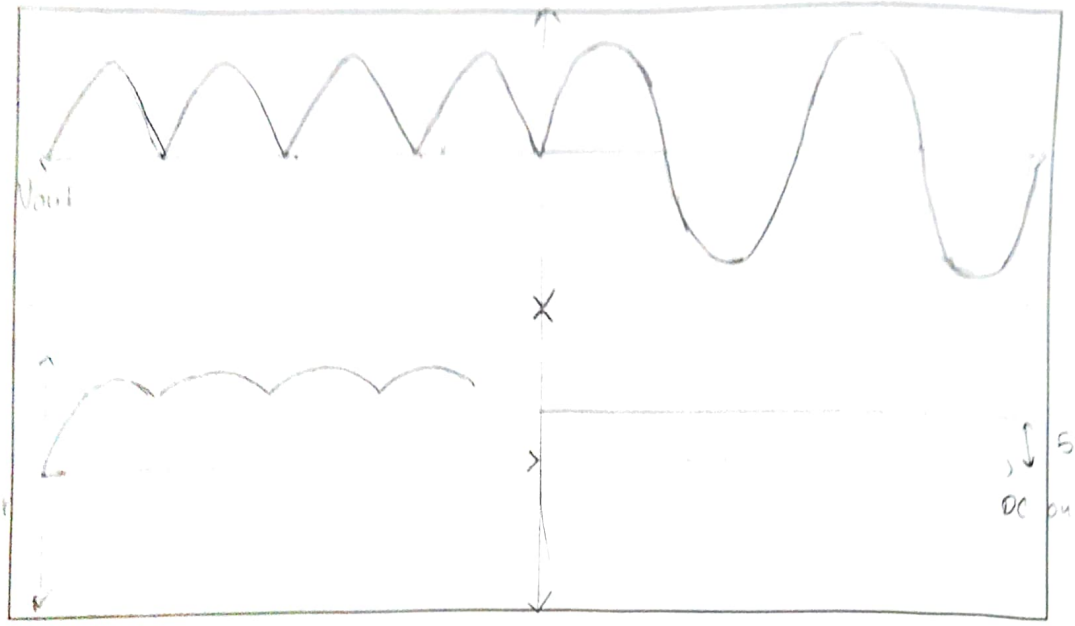


Circuit Diagram of mobile battery charger

Stepwise-Procedure:

1. Design circuit and connect it as shown in the circuit diagram using Proteus simulator.
2. Run the hardware and take scree shot of it to attach in the output.

Output waveforms observed on CRO:



Observation Table:

Vin(p&rms) (input of Rectifier in Volts)		Vout(peak) Output of Rectifier (in Volts)	DC output of 7805 (in Volts)
19.9	7.0	8 (with resistor)	5V
19.9	6.9		

Post Lab Subjective/Objective type Questions:

1. Explain working of Mobile Battery charger circuit
2. Explain the working principle of Turbo- charger for mobile phones .
3. State commonly used types of mobile phone batteries.
4. Explain how to maximize Battery Performance/ Battery life of your mobile phone?
5. Write important specifications of Voltage regulator IC 7805

Conclusion:

The mobile battery charger circuit was sucessfully implemented and its working was studied. The step-down transformar reduced the input AC voltage & the rectifirer and capacitor, converted AC to DC voltage. The IC7805 voltage regulator provided stable DC output of 5V. The output waveforms were studied.

Signature of faculty in-charge with Date:

Post Lab:

1. The transformer reduces the AC voltage to a lower value. Four diodes in the bridge rectifier convert AC into DC. The filter capacitor reduces ripples, and the IC 7805 produces a regulated 5V DC output.

2. A turbo charger, or fast charger, supplies high power to charge a phone faster. It uses efficient power conversion and communication b/w the charger and phone to control voltage and current charging speed depends on device compatibility, battery temp and charge level.

3 (i) Lithium-ion (Li-ion): Light weight, rechargable and has high energy density.
(ii) Lithium-polymer (Li-Po): Light weight, rechargable and suitable for slim mobile phone.
(iii) Nickel-metal hydride (Ni MH): Used mainly in older mobile phones.
(iv) Nickel-Cadmium (Ni Cd): Used in older phones but now uncommon.

4 (i) Avoid excessive heat and direct sunlight.
(ii) Use a compatible, good-quality charger.
(iii) Reduce screen brightness.
(iv) Close unnecessary background apps and enable battery saver.
(v) Used optimised charging features if available.

5. Type: Fixed positive voltage regulator
Output voltage: +5V DC
No. of terminals: 3 - Input, Ground and Output
Input voltage: Typically 7.5V or higher for proper regulation.
Max output current upto 1.5A for certain variants under suitable thermal conditions.
Protection: Thermal shutdown and current limiting
Application: Used in regulated DC power supplies.